

Power Devices

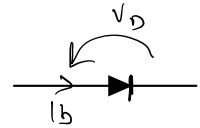
DIODES

Forward

$V_{0(t)}$
ON R

$$V_f : 0.6 - 0.7 V$$

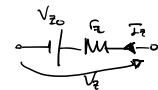
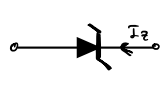
$$R_{on} : \frac{\Delta V_D}{\Delta I_D}$$



$$I_D(t) = I_S \left(e^{\frac{V_D(t)}{\eta V_T}} - 1 \right)$$

Zener

V_{BR}

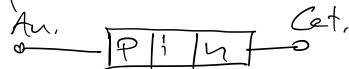


$V_Z = V_{Z0} + I_Z R_Z$
Used as ref. voltage

Reverse Recovery Time: t_{rr} : Forward $B \rightarrow$ Switch off
Carriers must be recomb
 $= t_s + t_{tr}$
Storage Time Reverse Trans.

Forward Recovery Time: t_{rf} : No charges to remove

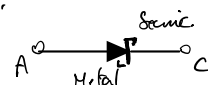
For power applications: P.I.N.



V_{BR} Increased
(400s V)
not typed

R_{on}, V_f incr.

Schottky Diode: Metal-Silicon



Fast switching (both dir)

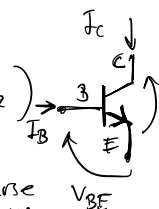
Lower R_{on}, V_f

High C in off

High reverse current I_S

Lower V_{BR} in abs val

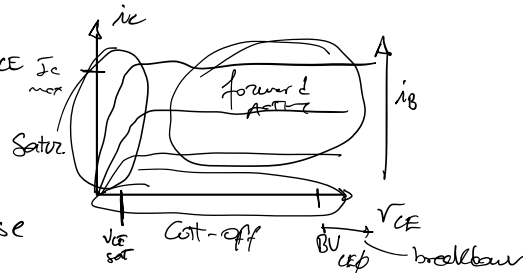
BJT (Bipolar Junction Transistor)



Cut-off: BE & BC reverse bias
 $\Rightarrow I_C = 0$

Forward: BE forward, BC reverse
Active $\Rightarrow$ Linear amplifier

Saturation: BE & BC forward Bias
 $\Rightarrow$ Closed switch $V_{CE} \approx 0$



Parameters: β : current gain
in forward: $I_C = \beta I_B$

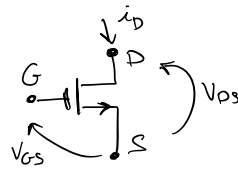
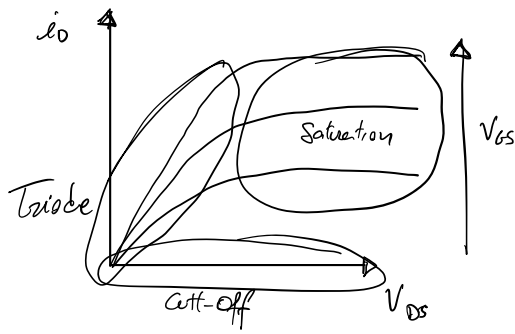
$V_{BE(on)}$: minimum V to turn on
Collect. Curr very small

$V_{CE(sat)}$: $\sim 0.2 - 0.3 V$

$V_{BE(sat)}$: When transistor in sat
 $\sim 0.7 V$

t_{rr} : Base-Collector junction problem
fwd $B \rightarrow$ need to recomb. charges

MOS Transistor (Metal-Oxide-Semicon.)



$R_{DS(on)}$ = slope of linear region ($V_{GS} \neq \phi$)

Threshold: V_{th} : minimum V to turn on $\sim 1-1.5V$

Cut-off: $V_{GS} < V_{th} \rightarrow i_D = 0$

Saturation: $V_{GS} > V_{th}$
 $V_{DS} > V_{GS} - V_{th}$ } Linear Amp

Triode: $V_{GS} > V_{th}$
 $V_{DS} < V_{GS} - V_{th}$ } Closed Switch
 $r_{DS} \approx \phi V$

Max DS voltage: BV_{DS}

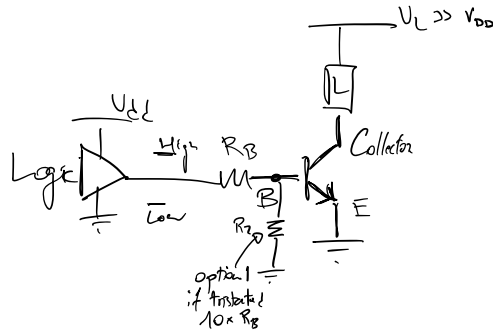
Max Drain Current: $I_{D,max}$

Max Power Dissip.: $P_{D,max} = I_D \cdot V_{DS}$

For POWER applications: DMOS

ACTUATOR DRIVING

Low Side, npn BST



Trans. OFF: force $V_{BE} < V_{BE(on)}$
 usually $V_{OL} < V_{BE(on)}$

Trans ON: $I_C = \beta I_B$
 To force saturation
 $I_B \gg \frac{I_C}{\beta}$
 $\beta_f = \frac{I_C}{I_B}$ (forced)
 $V_{CE} = V_{CE(sat)} \approx 0.2 \div 0.3V$
 $V_{BE} = V_{BE(sat)} \approx 0.8 \div 1V$

I_B depends on V_{in}
 $I_B = \frac{V_{in} - V_{BE(sat)}}{R_B}$
 To force Sat: $\gg \frac{I_C (V_{in} - V_{BE(sat)})}{\beta} \Rightarrow R_B \ll \frac{\beta (V_{in} - V_{BE(sat)})}{I_C}$

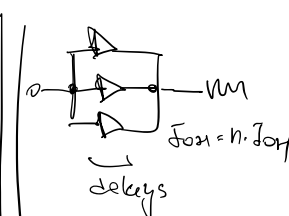
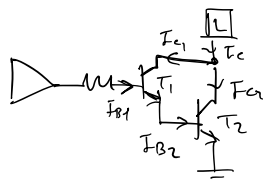
I_B should not exceed max I_{arr} of Digital = 0
 $\frac{V_{in} - V_{BE(sat)}}{R_B} < I_{OH} \Rightarrow R_B > \frac{V_{in} - V_{BE(sat)}}{I_{OH}}$

Current when Transistor Saturation: $I_C = \frac{V_L - V_{CE(sat)}}{R_L}$

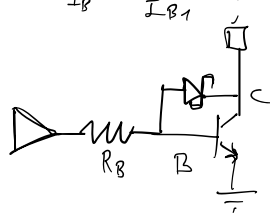
Higher Currents req.:

↳ Darlington Transistor

Increase β : $I_{B2} = \beta_1 I_{B1}$
 $I_{C2} = \beta_2 I_{B2} = \beta_2 \beta_1 I_{B1}$
 $I_C \approx I_{C2}$
 $\beta_{equiv} = \frac{I_C}{I_{B1}} = \frac{I_{C2}}{I_{B1}} = \beta_2 \beta_1$

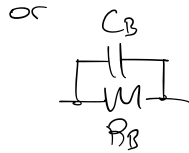


- Dynamic Beh.



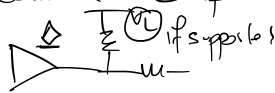
$$t_{err} = t_s + t_{rt}$$

Schottky to avoid saturation of BC junction
 $\Rightarrow$ no charges to remove



$\rightarrow$ Increase Base current during T_{on} $\rightarrow$ start c.
 After $\rightarrow$ open

- If Open Collector out



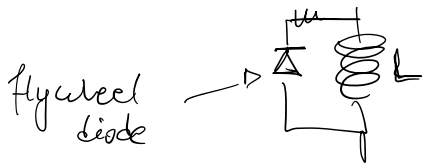
$$\frac{V_L}{R_{pu}} < I_{OL} \Rightarrow R_{pu} > \frac{V_L}{I_{OL}} \text{ to limit cur}$$

Lower R

When outp is High: R_{pu} series R_B R_{pu} should be low
 $R_{pu} + R_B < \frac{V_L - V_{BE_{sat}}}{I_C}$ $\rightarrow$ force sat

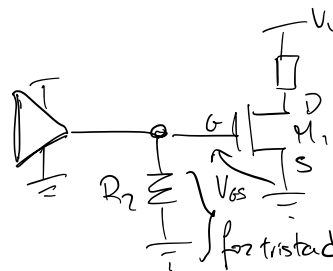
Upper

- Inductive Load



$$V(t) = L \frac{di(t)}{dt}$$

Low Side, nMOS



Transistor OFF: $V_{GS} < V_{th}$
 V_{OL} is fine ($< 1V$)

Transistor ON: Rise V_{GS} high enough $V_{GS} > V_{th}$ to bring in Triode
 The higher $V_{OV} = V_{GS} - V_{th} \rightarrow$ the lower V_{DS}
 the more in Triode

: Transfer char Linear (Triode & Sat.)

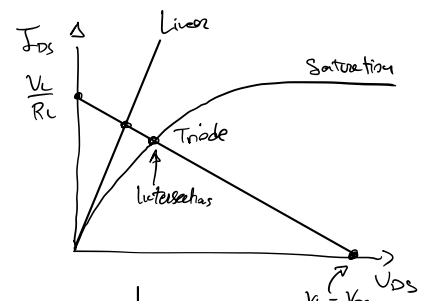
In all cases $V_{GS} > V_{th}$

$$\text{Linear: } I_{DS} = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th}) V_{DS}$$

$$R_{DS(on)} = \frac{V_{DS}}{I_{DS}} = \frac{1}{\mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})}$$

$$\text{Triode: } I_{DS} = \mu C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$\text{Saturation: } I_{DS} = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

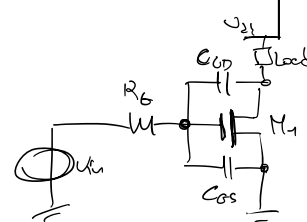
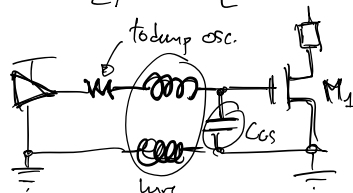


$$V_{DS} < V_{GS} - V_{th}$$

$$V_{DS} \geq V_{GS} - V_{th}$$

Intersect load line and V_{DS} char
 check that $V_{DS} < V_{GS} - V_{th}$
 For linear (resistor) load R_L

- Dynamic Beh.



C_{DS} amplified by Miller effect

Drive MOS with low outp. R (high f)
 $\rightarrow$ increase R

o High Side Driving, npn BST

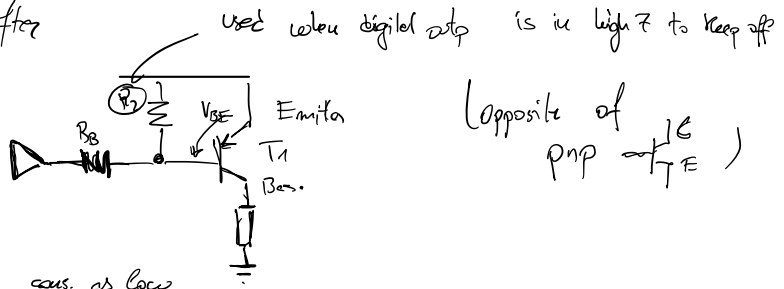
Solutions: npn cannot be switched ON usually

Use: - charge pump
- Level shifter

o High Side Driving, pnp BST

pnp often have lower β

2 cases o $V_{DD} = V_L \rightarrow$ same cons. as low side (just opposite sign)



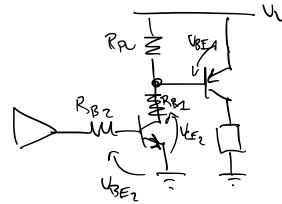
(opposite of $pnp \rightarrow \frac{I_C}{I_E}$)

o $V_{DD} < V_L$: Low logic outp switches trans. ON

: Trans can not be switched OFF! $V_{BE} = V_{DD} - V_L > V_{BE(on)}$ in Abs. val
negative

↳ use open collector and R_{PU} to V_L

↳ Use driving low npn T_2
 T_2 will enable same V_{as} but lower current



Logic Low $\rightarrow$ switch off low side npn T_2
High side pnp T_1 is also OFF

Logic High $\rightarrow$ saturates low side npn T_2
 $\Rightarrow$ current flows in the base of pnp T_1 and saturates

$$I_{B1} > \frac{I_{C1}}{\beta_1} \quad I_{B2} > \frac{I_{C2}}{\beta_2} = \frac{I_{B1}}{\beta_2}$$

$$R_{B2} < \frac{\beta_1 \beta_2}{I_{C1}} (V_{as} + V_{BE2sat})$$

$$R_{B2} > \frac{V_{as} - V_{BE2sat}}{I_{OH}} \quad \text{or} \quad R_{B2} > \frac{V_{DD} - V_{BE2sat}}{I_{OH}}$$

$$R_{PU} = \frac{V_{BE1sat}}{(10) \cdot I_{B2}}$$

o High Side, nMOS: difficult to Turn ON

$\rightarrow$ charge pump
 $\rightarrow$ level shifter

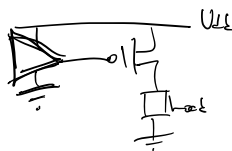
nMOS have lower $R_{DS(on)}$ comp to pMOS

- When load $V =$ to $V_{DD} \rightarrow$

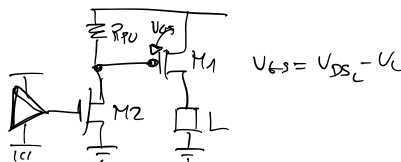
- When $V > V_{DD}$

High Side pMOS

Complementary to nMOS low side dr



When load V higher than what the digital circuit can sustain use low side nMOS



M_2 same volt. as M_1 but much lower current

P_{in} high $\rightarrow$ to switch off M_1 when M_2 off

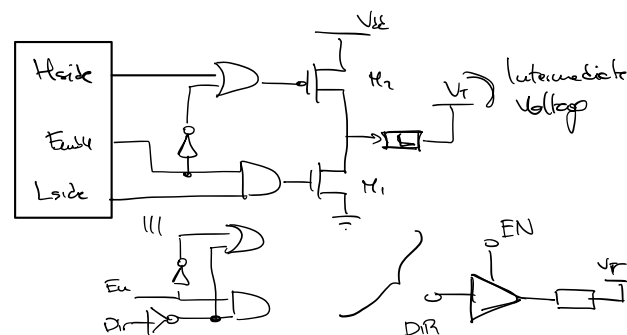
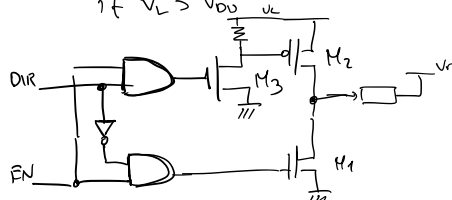
FLLOATING LOAD

Half H bridge

- Some loads need current in both dir.

- nMOS $\rightarrow$

- pMOS may require lowside nMOS if $V_L > V_{DD} - V_L$

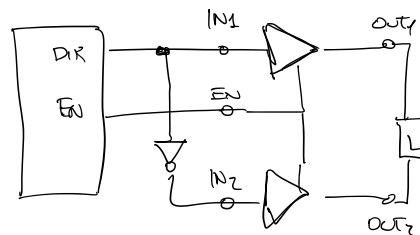
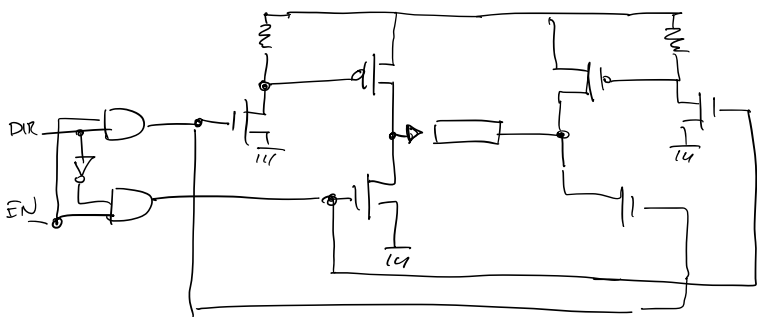


Full H bridge

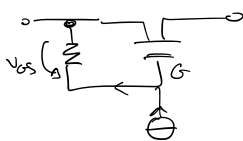
- Two half bridges

- Transistors always switch ON in opposite couples

- Load can be left floating = all transistors off



o Flating switch nmos



• POWER SUPPLY

Load Regulation:

$$S_I = R_{out} = \frac{\Delta V_{out}}{\Delta I_{out}}$$

$I_{dell} \Rightarrow$

Line Regulation

$$S_V = \frac{\Delta V_{out}}{\Delta V_{in}}$$

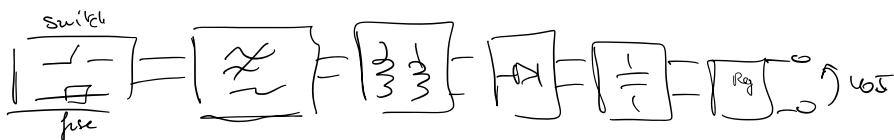
$I_{dell} \Rightarrow$

Efficiency

$$\eta = \frac{P_L}{P_{out}}$$

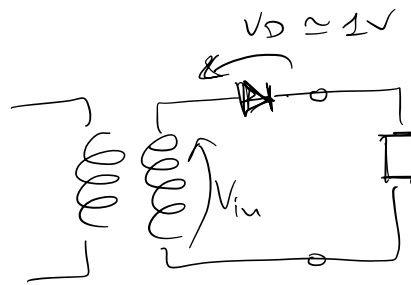
Linear reg. $\Rightarrow$ low eff.
+ Excellent sensitivity

Switching reg. $\Rightarrow$ high eff.
- Good sens

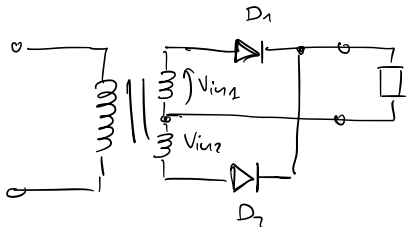


RECTIFIER

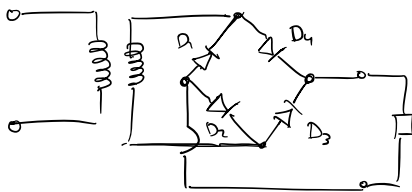
HALF WAVE



FULL WAVE



Diodes work separately $\Rightarrow$ drop $= V_D$
Current divided $\Rightarrow$ lower power diss.



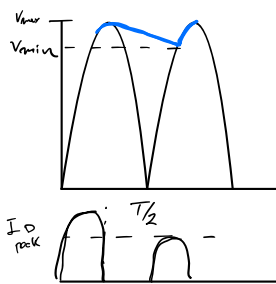
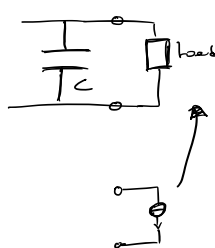
2 V_D Voltage drop
2 $\times$ Diodes $\rightarrow$ higher consumption
Normal transf. (simpler)

$\hookrightarrow$ Max REVERSE V must be seen.
Max CURRENT I_{avg}

FILTER

Convert pulsating output of rectifier into pseudo-DC

Simplest



$$V_{\text{max}} = V_A - V_D \quad \text{or} \quad V_{\text{max}} = V_A - 2V_D \quad \text{in 4 diodes config}$$

$$\Delta V_{\text{out}}(t) = \frac{I_{\text{load}} \cdot \tau}{C} \rightarrow \Delta V = \frac{I_{\text{load}} \cdot T}{C}$$

$$V_{\text{min}} = V_{\text{max}} - \Delta V$$

$$V_{\text{avg}} = \frac{V_{\text{max}} + V_{\text{min}}}{2} = \frac{V_{\text{max}} - \Delta V}{2}$$

For Half Rectifier : higher max $\Delta V = \frac{I_{\text{load}}}{C} \cdot T$

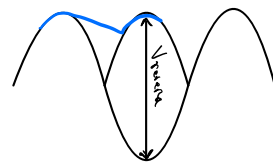
$$V_{\text{avg}} = V_{\text{max}} - \frac{\Delta V}{2}$$

Longer conduction time but higher peak

MAX REVERSE VOLTAGE

When diode not conducting:

- $V_{max} \approx V_A$
- Input voltage goes to min: $-V_A$
- Total Reverse voltage is approx $2V_A$
Peak-to-Peak voltage @ o/p of transformer



For Half Wave & Full Wave (2 Diodes): choose diodes with $V_{BR} > 2V_A$

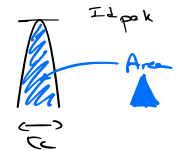
For Full Wave (4 diodes) $V_{BR} > V_A$ since 2 diodes in series

CURRENT PEAK

$$V_{in} = V_{max} \cos\left(\frac{2\pi}{T} t\right)$$

$$t_c = \frac{T}{2\pi} \cos^{-1}\left(\frac{V_{max} - \Delta V}{V_{max}}\right)$$

where t_c base I_{Dpeak} height



Conducting time t_c longer the higher the residual Ripple ΔV

$$\text{HW: } \frac{t_c I_{Dpeak}}{2} = T I_{out} \Rightarrow I_{Dpeak} = 2 I_{out} \frac{T}{t_c}$$

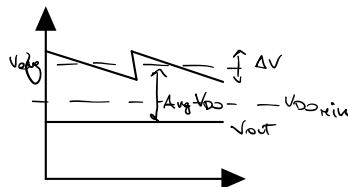
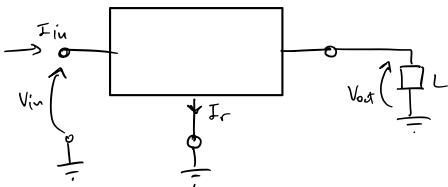
$$\text{FW: } \frac{t_c I_{Dpeak}}{2} = \frac{T}{2} I_{out} \Rightarrow I_{Dpeak} = I_{out} \frac{T}{t_c}$$

trade off with ripple



Higher $C \rightarrow$ lower ripple

LINEAR REGULATOR



$$I_{in} = I_R + I_{out} \quad \text{Typ. } I_R \ll I_{out}$$

V_{DS} should be $\geq V_{DSmin}$

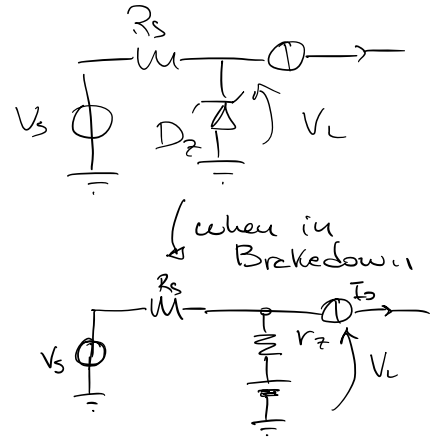
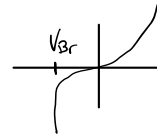
$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_o \cdot I_o}{V_i \cdot I_i} \approx \frac{V_{out}}{V_{in}}$$

$$\text{on Avg: } \eta \approx \frac{V_o}{V_i} = \frac{V_{avg} - V_{DS}}{V_{avg}} \approx 1 - \frac{V_{DS}}{V_{avg}}$$

$$\eta_{max} = 1 - \frac{\frac{\Delta V}{2} + V_{DSmin}}{V_{avg}}$$

ZENER LINEAR REGULATOR

$$V_L = \frac{V_S + \frac{V_{Z0}}{r_z} - I_0}{\frac{1}{R_S} + \frac{1}{r_z}}$$



Minimum Zener current : worst case $\circ V_{S \min} = V_{avg} - \frac{\Delta V}{2}$
 $\circ I_0 = I_{Z \min}$

$$\Rightarrow I_Z = \frac{V_L' - V_{Z0}}{r_z} > I_{Z \min} \Rightarrow V_L' = \frac{\frac{V_{S \min} + V_{Z0} - I_{Z \min}}{R_S} + \frac{V_{Z0}}{r_z}}{\frac{1}{R_S} + \frac{1}{r_z}}$$

$$\rightarrow R_S < R_{S \min} = \frac{V_{S \min} - V_{Z0} - r_z I_{Z \min}}{I_{Z \min} + I_{Z \max}}$$

Maximum Zener Power : worst case $\circ V_{S \max} = V_{avg} + \frac{\Delta V}{2}$
 $\circ I_0 = I_{Z \min}$ or (worse) $\circ I_0 = I_{Z \max}$

$$\Rightarrow I_Z V_L'' = \frac{V_L'' - V_{Z0}}{r_z} V_L'' < P_{Z \max}$$

$$V_L'' = \frac{\frac{V_{S \max}}{R_S} + \frac{V_{Z0}}{r_z}}{\frac{1}{R_S} + \frac{1}{r_z}}$$

LOAD REGULATION:

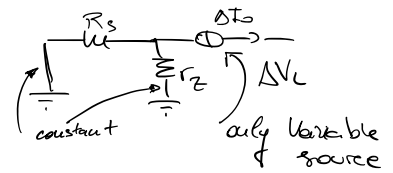
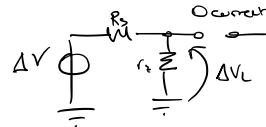
$$S_L = \frac{\Delta V_L}{\Delta I_0} = \frac{R_S r_z}{R_S + r_z} = R_S \parallel r_z \approx r_z$$

Lower the better

LINE REGULATION:

$$S_V = \frac{\Delta V_L}{\Delta V} = \frac{r_z}{R_S + r_z}$$

Higher the better



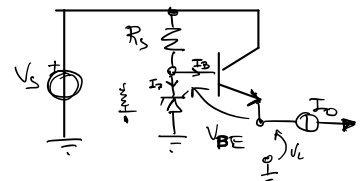
We used higher R_S but we have constraints on output current $\Rightarrow$ BJT

$$V_L = V_D - V_{BE} \leq V_D - 0.7V$$

R_S can be much bigger $\Rightarrow R_S < R_{S \max} = \frac{V_{S \min} - V_{Z0} - r_z I_{Z \min}}{I_{Z \max} - I_{Z \min}}$

Voltage @ V_{BE}
 constr. on min current

The output Res. of emitter = $\frac{R_{S \text{ seen from base}}}{\beta}$

$$S_I = \frac{1}{\beta} \cdot \frac{R_S r_z}{R_S + r_z}$$


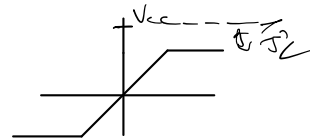
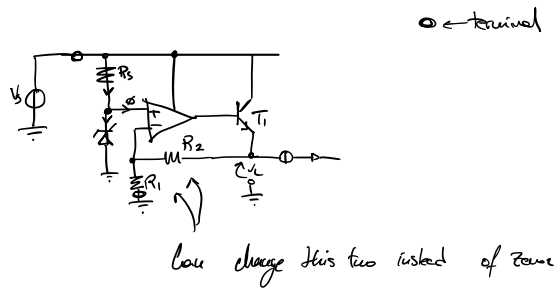
3 Terminal Regulator

-OA with negative feedback

$$V_L = V_{ref} = \left(1 + \frac{R_2}{R_1}\right)$$

$$R_S < R_{S_{max}} = \frac{V_{S_{min}} - V_{L0} - r_z I_{L_{min}}}{I_{L_{min}}}$$

Load regulation better is OpAmp has very low output R



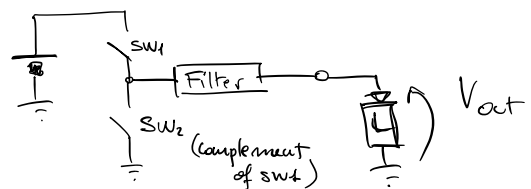
$V_S > V_L$ by amount = drop from voltage supply $\rightarrow$ output OpAmp
additional amount = V_{BE} of output PNP.

$$V_{D_{min}} \leq 2 \div 3V$$

$\rightarrow$ Linear: very very good output but low efficiency

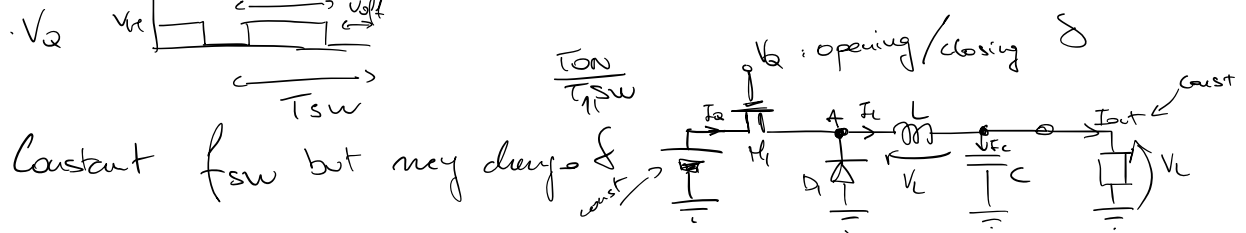
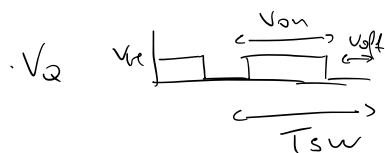
SWITCHING POWER SUPPLY

"Transformer" for DC
Control with high switching for low



(Bad line regulation)

BUCK (STEP DOWN) DC-DC



$$V_{LAVG} = 0V$$

$$I_{CAVG} = 0A$$

delta not absol. val.

During T_{ON} : current in L $\Delta I_{LON} = \frac{V_L T_{ON}}{L} = \frac{1}{L} (V_{in} - V_{out}) T_{ON}$
 current in M_1 $I_Q = I_L$
 current in D_1 $I_D = 0$

During T_{OFF} : $V_A = 0$ (neglect voltage drop on D_1)
 $V_L = 0 - V_{out} = -V_{out}$ (const)

$$\Delta I_{LOFF} = \frac{V_L T_{OFF}}{L} = -\frac{1}{L} V_{out} T_{OFF}$$

$$I_Q = 0 \text{ in } M_1$$

$$I_D = I_L$$

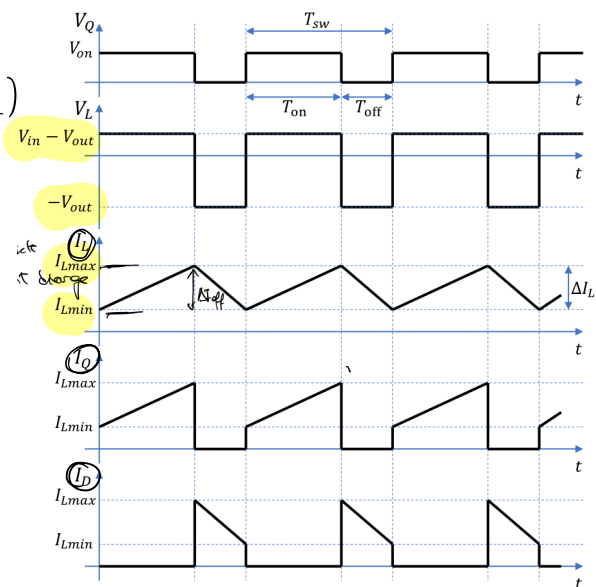
$$\Delta I_{LON} + \Delta I_{LOFF} = 0 \Rightarrow V_{out} = \delta V_{in}$$

$$\frac{1}{L} (V_{in} - V_{out}) T_{ON} - \frac{1}{L} V_{out} T_{OFF} = 0$$

always lower

$$I_{in} = I_Q \text{ since } P_{in} = P_{out} \Rightarrow I_{inavg} = \delta I_{out}$$

$$I_{ar} = \frac{I_{Lmax} + I_{Lmin}}{2}$$



Current Rating for Inductor: either compute I_{Lmax} or $I_{Lrms} = \frac{2}{\sqrt{3}} I_{ar}$

Current Rating for Capacitor: max is $\frac{\Delta I_L}{2}$ or $I_{Crms} = \frac{1}{\sqrt{3}} I_{ar}$ (Half of Ind)

LINE REGULATION: outp. v depends on inp. v. : $S_r = \frac{\Delta V_{out}}{\Delta V_{in}} = \delta$

It's a DC-DC converter, not regul.

→ need negative feedback circuit that controls δ

LOAD REGULATION: Outp. voltage does not depend on I_{out}

$$S_I = 0$$

without considering parasitic R_s , @ steady state
 during transients $V_{out} \neq \delta V_{in}$
 Valid only in CCM

(BUCK) DISCONTINUOUS CURR. MODE

- Current in L reaches 0 $\rightarrow I_{Lmin} = 0$, Current change in T_{on} remains the same
- Current goes to 0 during T_2 $I_{Lmax} = \frac{1}{L} V_{out} T_2 \Rightarrow T_2 = \frac{V_{in} - V_{out}}{V_{out}} T_{on}$

Now V_{out} depends on I_{out} $V_{out} = \frac{V_{in}}{\frac{2LI_{out}}{\delta^2 I_{sw} V_{in}} + 1}$ Depends on $\frac{L}{I_{sw} I_{out}} \left. \vphantom{\frac{L}{I_{sw} I_{out}}} \right\} S_I > 0!$

$$I_{DCH} = \frac{\Delta I_{LON}}{2} = \frac{(1-\delta)V_{out}}{2L f_{sw}}$$

Without regulator: outp V_{out} increases for low I_{out}
for $I_{out} = 0 \rightarrow V_{out} = V_{in}$ regardless of δ
Given $I_{outmin} \rightarrow$ Minimum L to have CCM is $L > \frac{(1-\delta)V_{out}}{2I_{outmin} f_{sw}}$

For worst case $\delta=0$ $L > \frac{V_{out}}{2I_{outmin} f_{sw}}$

To guarantee: $\frac{I_{out}}{4C f_{sw}} < \Delta V_{out} \rightarrow C > \frac{I_{out}}{4\Delta V_{out} f_{sw}}$
RIPPLE

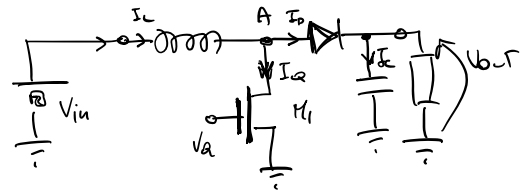
$$\Delta V_{out} = \frac{1}{C} \frac{1}{2} \frac{I_{sw}}{2} \frac{\Delta I_L}{2} = \frac{\Delta I_L}{8C f_{sw}} \quad \text{Worst case } \Delta I_L = 2I_{out}$$

	MOS	Diode
Current	$\frac{2}{\sqrt{3}} I_{out}$	$\frac{2}{\sqrt{3}} I_{out}$
Voltage	V_{in}	V_{in}

	Inductor	Capacitor
Current	$\frac{2}{\sqrt{3}} I_{out}$	$\frac{1}{\sqrt{3}} I_{out}$
Voltage	V_{in}	V_{out}

BOOST (STEP UP) DC-DC CASE

T_{on} : M_1 conducting $\rightarrow V_A = 0$
 D_1 reverse bias
 $V_L = V_{in}$: Current in L changes linearly
 $\Delta I_{LON} = \frac{V_L T_{on}}{L} = \frac{1}{L} V_{in} T_{on}$
 M_1 : $I_{M1} = I_L$
 D_1 : $I_{D1} = 0$



T_{off} : M_1 cut off
 D_1 forced conduction: $V_A = V_{out}$
 $V_L = V_{in} - V_{out}$: $\Delta I_{Loff} = \frac{V_L T_{off}}{L} = \frac{1}{L} (V_{in} - V_{out}) T_{off}$
 $I_{M1} = 0$
 $I_{D1} = I_L$

with $V_{in} > 0$, $V_{out} > 0$, $V_{out} > V_{in}$

$$\Delta I_{LON} + \Delta I_{Loff} = 0 \Rightarrow \frac{1}{L} V_{in} T_{on} + \frac{1}{L} (V_{in} - V_{out}) T_{off} = 0 \Rightarrow \frac{V_{out}}{V_{in}} = \frac{1}{1-\delta}$$

Current in L = Current in D
in CCM ($I_{Lmin} > 0$)

$$I_{Lmax} = \frac{1}{1-\delta} I_{out} + \frac{\Delta I_L}{2}$$

$$I_{Lmin} = \frac{1}{1-\delta} I_{out} - \frac{\Delta I_L}{2}$$

Current from voltage source

$$I_{in} = I_L \leftarrow \text{clearer}$$

$$P_{out} = P_{in} \rightarrow I_{avg} = I_{Lavg} = \frac{V_{out}}{V_{in}} \cdot I_{out} = \frac{1}{1-\delta} I_{out}$$

CURRENT RATINGS

$$L: I_{Lmax} \text{ or } I_{Lrms} = \frac{2}{\sqrt{3}} I_{Lavg} = \frac{2}{3} \frac{1}{1-\delta} I_{out}$$

$$C: \text{When low } \delta \text{ worst } I_{Crms} = I_{out} \quad \Delta V_{out} = \frac{I_{out}}{C f_{sw}} = \frac{I_{out} T_{sw}}{C}$$

LINE REGULATION:

$$S_V = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{1}{1-\delta} \quad \text{Even worse than Buck} \rightarrow \text{needs regulator}$$

LOAD REGULATOR: Outp. Voltage does not dep on Outp Current I_{out}
 $S_I = 0$

CCM for $L > \frac{V_{in}}{2 I_{out} f_{sw}}$

DCM: Current during T_{on} same

$$\text{Current goes to 0 during } T_2: I_{Lmax} = -\frac{1}{L} (V_{in} - V_{out}) T_2 \rightarrow T_2 = -\frac{V_{in}}{V_{in} - V_{out}} T_{on}$$

$$I_L \text{ still equal to } \frac{1}{2} (T_{on} + T_2) I_{Lmax} = \frac{V_{out}}{V_{in}} I_{out} T_{sw}$$

$$V_{out} = V_{in} \left(\frac{\delta^2 T_{sw} V_{in}}{2 L I_{out}} + 1 \right) \quad S_I > 0$$

BUCK-BOOST

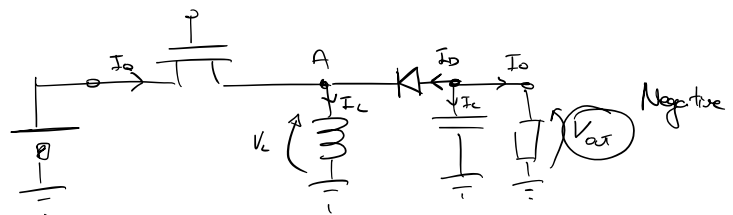
$$T_{on}: V_A = V_{in}$$

D_1 Reverse Bias

$$\Delta I_{Lon} = \frac{V_L T_{on}}{L} = \frac{1}{L} V_{in} T_{on}$$

$$I_{Q} = I_L$$

$$I_D = 0$$



$T_{off}: D_1$ forced conduction

$$\rightarrow V_A = V_{out}$$

$$V_L = V_{out}$$

$$\Delta I_{Loff} = \frac{1}{L} V_{out} T_{off}$$

$$I_Q = 0$$

$$I_D = I_L$$

Assume $V_{in} > 0$ $V_{out} < 0$

$$\Delta I_{L_{on}} + \Delta I_{L_{off}} = 0$$

$$\frac{1}{L} V_{out} T_{on} + \frac{1}{L} V_{out} T_{off} = 0$$

$$\rightarrow \frac{V_{out}}{V_{in}} = -\frac{T_{on}}{T_{off}} = -\frac{\delta}{1-\delta}$$

$$V_{out} = -\frac{\delta}{1-\delta} V_{in}$$

negative! $- \infty < V_{out} < 0$

Current in L : Avg curr in Load & in diode should be identical in ABS value because $I_{c, avg} = 0$

$$I_{L_{max}} = -\frac{1}{1-\delta} I_{out} + \frac{\Delta I_L}{2}$$

$$I_{L_{min}} = -\frac{1}{1-\delta} I_{out} - \frac{\Delta I_L}{2}$$

Output current is Pulsated

If δ high $\rightarrow$ Voltage source almost shorted to GND

DCM mode

$$I_{L_{on}} \text{ slope} = \frac{1}{L} V_{in} T_{on} \quad I_{L_{max}} = \Delta I_{L_{on}} = \frac{1}{L} V_{in} T_{on}$$

$$\text{During } T_2 \quad I_{L_{max}} = -\frac{1}{L} V_{out} T_2 \Rightarrow T_2 = -\frac{V_{in}}{V_{out}} T_{on}$$

$$\text{On the avg current in } D = I_{out} \rightarrow \frac{1}{2} T_2 I_{L_{max}} = I_{out} T_{sw}$$

$$V_{out} = -\frac{\delta^2 T_{sw} V_{in}}{2 L I_{out}} \Rightarrow S_F > 0$$

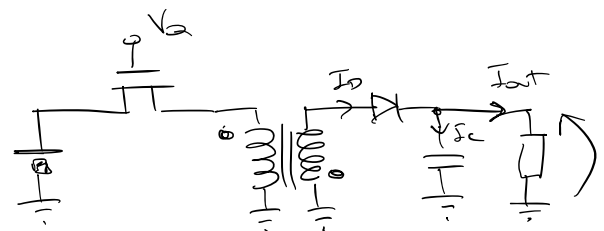
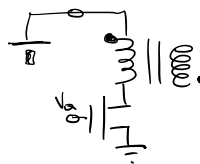
FLYBACK DC-DC converter

• $V_{out} > 0$

$$V_{out} = \frac{\delta}{1-\delta} V_{in}$$

• Transformer works @ f_{sw} , can have multiple secondaries

• Better to Low-Side - Drive



REGULATOR CIRCUIT

$$\beta = \frac{R_1}{R_1 + R_2}$$

ϵ : error

$$\delta = H V_E \quad H = \frac{1}{V_{KPP}}$$

$$T_{ON} = T_{SW} = V_E : V_{KPP}$$

$$V_{out} = \delta V_{in} = \frac{V_E}{V_{KPP}} V_{in} = \frac{G \epsilon}{V_{KPP}} V_{in}$$

$$= \frac{G}{V_{KPP}} (V_r - \beta V_{out}) V_{in}$$

$$= \frac{G}{V_{KPP}} \left(V_r - \frac{R_1}{R_1 + R_2} V_{out} \right) V_{in}$$

$$= \frac{1}{\beta} \cdot \frac{A \beta}{1 + A \beta} V_r \quad \text{with} \quad A = \frac{G V_{in}}{V_{KPP}}$$

$$\lim_{A \beta \rightarrow \infty} V_{out} = \frac{1}{\beta} V_r = \left(1 + \frac{R_2}{R_1} \right) V_r$$

As long as $0 < V_{out} < V_{in}$

for boost $V_{out} > V_{in}$
for buck-boost $V_{out} < 0$

